OPC UA

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COURSE OVERVIEW — CHAPTER BY CHAPTER

1. **OPC UA Architecture** — A committee of every major automation vendor built OPC UA intentionally as a vendor-neutral replacement for OPC Classic. This chapter explains the unified architecture model, the death of DCOM, and why client/server and pub/sub coexist in the same standard.
2. **Information Model** — OPC UA's address space is a graph of nodes and references. Engineers learn NodeID types, object hierarchies, and how a client can browse an unknown server and understand its entire data structure without a custom driver.
3. **Services** — Read, Write, Browse, Subscribe, and CreateMonitoredItems replace what previously required custom OPC Classic drivers for every vendor. This chapter covers the full service set and which services matter most in production.
4. **Sessions & Channels** — SecureChannel establishes the encrypted transport. Session sits on top and carries the user context. This chapter covers token expiration, reconnect behavior, and the distinction between channel failure and session failure.
5. **Security** — Security Mode None is the production default at most sites because commissioning happened under a deadline. This chapter covers Modes Sign and SignAndEncrypt, certificate exchange, PKI infrastructure, and user authentication — and why skipping them costs more than implementing them.
6. **Subscriptions** — Publishing interval, sampling interval, deadband filters, queue size, and MonitoredItem configuration determine how much bandwidth an OPC UA connection consumes. Engineers learn to tune subscriptions instead of flooding historians with redundant data.
7. **Data Types** — Built-in scalars, structured types, enumerations, multi-dimensional arrays, and QualityStatusCode interpretation under partial network failure. The data type system is what allows OPC UA to represent any process value without custom encoding.
8. **Discovery** — Local Discovery Server and Global Discovery Server allow clients to find OPC UA servers without prior configuration. This chapter covers endpoint enumeration and the security implications of an open discovery infrastructure.
9. **Implementation** — Selecting a stack (open62541 vs. commercial), configuring Ignition, Siemens, and Allen-Bradley OPC UA servers, and the common integration patterns that appear in every multi-vendor plant network.
10. **Protocol Lab** — Wireshark OPC UA session analysis, security mode verification from a live capture, and subscription traffic profiling. Engineers leave this chapter able to prove what their OPC UA configuration is actually doing on the wire.

Certification Relevance

- **OPC Foundation Certified** — Developer and Integrator exams test the information model, service set, and security configuration directly covered in this course.
- **GICSP (GIAC)** — OPC UA security modes and integration patterns appear in ICS security exam prep domains.
- **ISA CCST** — Industrial networking and system integration standards are tested at all CCST levels.

Next Steps

1. Complete OPC UA alongside the Security guide for maximum overlap with GICSP exam domains.
2. Download open62541 (free, open-source C stack) and run a local OPC UA server against UA Expert.
3. Review OPC Foundation spec Parts 1 (Concepts) and 4 (Services) for the normative definitions.
4. Apply at opcfoundation.org for the OPC UA Integrator certification exam.